

NAME: _____ DATE: _____

6.5-1 Mass, Mole, and Entity

Calculate the number of molecules present in each of the following samples.

1. 4.56 mol of ethyne, C_2H_2

2. 2.6 kg of propane, C_3H_8

Calculate the amount of substance (in moles) present in each of the following samples.

3. 4.62×10^{24} molecules of oxygen, O_2

4. 2.3×10^{22} atoms of neon, Ne

5. 1.33×10^{25} formula units of sodium chloride, NaCl

Calculate the number of hydrogen atoms in each of the following samples.

6. 3.4 mol of hexane, C_6H_{14}

7. 45.7 g of octane, C_8H_{18}

8. 6.23×10^4 g of ammonium nitrate, NH_4NO_3

9. 6.23 mg of sucrose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$